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PII: S2451-9022(25)00015-1

DOI: <https://doi.org/10.1016/j.bpsc.2024.12.016>

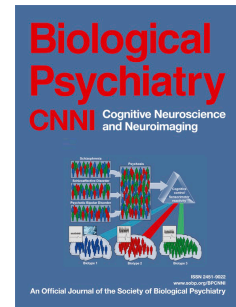
Reference: BPSC 1369

To appear in: *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*

Received Date: 18 April 2024

Revised Date: 29 December 2024

Accepted Date: 30 December 2024



Please cite this article as: Tripathi V., Batta I., Zamani A., Atad D.A., Sheth S.K.S., Zhang J., Wager T.D., Whitfield-Gabrieli S., Uddin L.Q., Prakash R.S. & Bauer C.C.C., Default mode network functional connectivity as a transdiagnostic biomarker of cognitive function, *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* (2025), doi: <https://doi.org/10.1016/j.bpsc.2024.12.016>.

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# Title

Default mode network functional connectivity as a transdiagnostic biomarker of cognitive function

Short title: DMN functional connectivity as a cognitive biomarker

# Authors

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## Abstract

The default mode network (DMN) is intricately linked with processes such as self-referential thinking, episodic memory recall, goal-directed cognition, self-projection, and theory of mind. Over recent years, there has been a surge in examining its functional connectivity, particularly its relationship with frontoparietal networks (FPN) involved in top-down attention, executive function, and cognitive control. The fluidity in switching between these internal and external modes of processing—highlighted by anti-correlated functional connectivity—has been proposed as an indicator of cognitive health. Due to the ease of estimation of functional connectivity-based measures through resting state fMRI paradigms, there is now a wealth of large-scale datasets, paving the way for standardized connectivity benchmarks. This review delves into the promising role of DMN connectivity metrics as potential biomarkers of cognitive state across attention, internal mentation, mind wandering and meditation states, and investigating deviations in trait-level measures across aging and in clinical conditions such as Alzheimer's disease, Parkinson's, depression, ADHD, and others. Additionally, we tackle the issue of reliability of network estimation and functional connectivity and share recommendations for using functional connectivity measures as a biomarker of cognitive health.

## Introduction

The default mode network (DMN) is a macroscale functional brain network associated with higher-order and internally-oriented cognitive processes including episodic projection, theory of mind (TOM), and autobiographical processing (1). Recent studies have also highlighted its role in goal directed self-generated thoughts, semantic processing and episodic memory retrieval (2,3). Its constituent brain regions span the posterior cingulate cortex (PCC), medial prefrontal cortex (mPFC), temporo-parietal junction (TPJ), angular gyrus, inferior parietal lobule (IPL) and the temporal cortex (4–6). The DMN can be decomposed into multiple subnetworks (7,8) where one subnetwork is associated with episodic projection (thinking about the past or the future) and

another with thinking about others/TOM (9,10). DMN activity is highly task- and context-dependent and changes across days and scanning sessions (11,12). Recently, the use of precision imaging, where subjects are scanned across multiple sessions, has illustrated the variability of DMN and other higher order cognitive networks across individuals (13–15) suggesting that averaged atlases may lose a lot of individual-specific variations and may result in less reliable network estimations (16,17).

The relationship between the DMN and other large scale brain networks is of growing interest to the neuroimaging community. While inconsistencies surrounding network nomenclature hamper progress in this area (18,19), reproducible findings suggest the DMN frequently interacts with multiple other networks. The frontoparietal network (FPN), also known as the cognitive control network (CCN) or Central Executive Network (CEN)(18), is anchored predominantly across the lateral prefrontal cortex, dorsal IPL, and supplementary motor area and is involved in top down control processes like executive function and working memory. These regions often co-activate with the dorsal attention network (DAN), which is centered in the posterior parietal cortex (PPC) along with the frontal eye fields (FEF) in the PFC. The Salience Network (SAL), comprising the anterior cingulate cortex (ACC) and anterior insular cortex (AI), is involved in deploying attentional resources and is associated with switching between internal and external attention (20). Though traditionally considered exclusive to externally-oriented tasks and displaying an antagonistic relationship with DMN, the FPN is now understood to also support internally-oriented cognition as evidenced by its co-activation with the DMN during goal-directed thought (2,21–23).

DMN anti-correlation refers to the phenomenon where activity in the DMN decreases when activity in task-positive networks, such as the FPN, increases during goal-directed tasks (24). This inverse relationship reflects the DMN's involvement in self-referential or internally focused processes, in contrast to externally directed attention. Early studies reported negative

1 correlations between a node of the DMN (PCC) and lateral prefrontal cortices (5). Research  
2 provided further evidence of negative correlations (or anti-correlations) between the DMN and  
3 other frontoparietal large-scale brain networks (4,25). Studies since have shown that individual  
4 differences in this DMN connectivity is associated with a wide array of behavioral variability  
5 (12,26,27). Heterogeneity within these networks and how they interact across different cognitive  
6 states and time has also been explored (12,28,29).

7  
8 In this review, we highlight the literature on the DMN functional connectivity (FC) with different  
9 networks across various cognitive states like attention, mind-wandering, meditation and goal  
10 directed cognition and explore trait-level changes in FC. We examine if DMN FC can be used as  
11 a stable biomarker of cognition and health. We conclude with guidelines and recommendations  
12 of how to estimate FC measures across these networks for future use in precision psychiatry.  
13

## 14 State Level Changes

15 First, we describe DMN FC across various cognitive states from the extensive neuroimaging  
16 literature on healthy participants.  
17

### 18 DMN and attention/mind-wandering

19  
20 Although attention is often considered an elusive construct because of its centrality in nearly all  
21 aspects of our waking lives (30), it is frequently defined as involving processes that select and  
22 prioritize representations most relevant to the task at hand, while filtering out task-irrelevant  
23 representations (31). Individual differences in the strength of anti-correlation between the DMN  
24 and FPN during attention task performance are correlated with response time variability.

Greater network segregation, measured as stronger negative correlations between the DMN and FPN, is associated with less variable behavioral performance on a flanker attention task (26). These findings have since been directly replicated in a large population neuroscience dataset, where stronger negative correlations have been further linked with reduced attention problems (32).

Mind-wandering, a phenomenological construct often considered to be antithetical to focused attention, involves the spontaneous shifting of focused attention away from the external environment to inner thoughts (33–35). Seminal neuroimaging studies investigating attention and mind-wandering traditionally linked activity in the FPN to supporting selective and sustained attention (36), and, in contrast, activity within the DMN to mind-wandering (36,37). Studies have used a rest condition where participants were instructed to do “nothing” to identify the DMN (1,6). During these rest periods, blood flow in midline cortical structures increased, suggesting the link between neuronal activity in DMN regions and spontaneous cognition or mind-wandering. Examining this link directly, Mason and colleagues (37) compared neural activations when participants performed practiced versus novel task sequences. Increases in automaticity were associated with a greater predilection for task-unrelated thoughts and less reduction in regions of the DMN compared with a novel task, highlighting the involvement of the network during tasks that result in greater off-task thinking. Furthermore, a meta-analytic review synthesizing 24 neuroimaging investigations of mind-wandering solidified the recognition of the key role played by the transmodal nodes of the DMN (specifically the PCC/precuneus, the mPFC, and the coupling between nodes of the DMN and FPN) in supporting the internal stream of thought (38).

Recent use of connectome-based predictive modeling (CPM) has found a whole-brain signature of sustained attention which involves multiple canonical networks beyond the FPN and DMN,

including the visual network, the cerebellar network, and the motor network (39,40). In related CPM research, a whole-brain model of mind-wandering was developed (41) which provided support for the differential involvement of the key canonical networks, including the DMN, the somatomotor network, the DAN, the ventral attention network, the visual network, and the FPN, in both high and low mind-wandering. Also, DMN anticorrelations were highly predictive of within subject variation of mind wandering (42)

Collectively, while earlier studies on attention and mind-wandering supported the double dissociation of FPN and DMN in the seemingly opposing phenomena of attention and mind-wandering, more recent empirical support highlights the involvement of both these networks, along with others , in supporting attention and mind-wandering.

## Dynamic interactions between DMN and FPN

A recent line of research using carefully designed task paradigms related to semantic processing, retrieval of episodic memory, and goal-directed cognition has shown that the DMN and FPN regions can act in conjunction towards these tasks rather than in an antagonistic manner (2,3,43). Even parts of the DMN are recruited during some executive processing like task switching (23). FPN can track trial-to-trial difficulty in internal mentation paradigms like episodic projection and scene construction (44). Studies investigating how creative processes involve various networks have found a dynamic interplay between DN, FPN and SAL during idea generation and evaluation processes. During idea generation, which is a bottom up process, DMN is anticorrelated with FPN but during the idea evaluation process the correlation increases between the networks highlighting cooperation to filter out ideas. Overall these studies highlight that goal directed self generated thought involves the use of both DMN and FPN (45–47). It could be possible that subnetworks within FPN specialize in both their function and connectivity with a subnetwork being more functional correlated with DAN whereas another

with DMN (28). Using precisely defined networks within deeply scanned individuals (16) we might be able to discern the functional specialization within the FPN and its interaction with DMN.

## DMN and meditation

Meditation refers to contemplative practices that involve regulating both body and mind through cultivating a state of heightened awareness (48) and includes a wide variety of techniques and approaches rooted in various traditions (49). Recent years have seen a surge of meditation and mindfulness-based interventions carried out either as therapeutic tools or interventions for the general public (50,51). Indeed, a large body of ongoing work converges on the positive effects of meditation, including enhanced emotion regulation, wellbeing and stress reduction (52–54), enhanced attentional skills (55,56), and alleviation of the symptoms of various mental conditions. Despite attempts to categorize diverse meditation styles (57–59), the neural correlates of meditation remain complex to describe simply, often exhibiting discrepancies across different styles (60,61).

Nevertheless, converging evidence based on study of various practices points toward the DMN as a key network whose activity and FC is modified both during the meditative state and as a trait following regular meditation practice (62,63). Numerous studies have demonstrated reduced activity in various nodes of the DMN during meditative states compared to rest (63–65) and also beyond the de-activation induced by task (66). Reduced DMN activity has also been shown during resting state of experienced meditators compared to controls (67,68).

Furthermore, FC within the DMN is altered during and following meditation. Evidence for decreased FC between DMN nodes involved in mind-wandering and self-referential processes has been observed in experienced meditators compared to controls during meditation (64,65,69) and at rest (67,68,70). Conversely, evidence for increased within-DMN FC has been



1 observed during meditation (67) and at rest, specifically in the mPFC (69,71). Despite these  
2 heterogeneous results, the emerging trend points towards a decoupling of regular DMN  
3 configuration and activation patterns in experienced meditators, suggesting a shift towards a  
4 mode less prone to mind-wandering and self-referential thinking, and more rooted in present-  
5 moment awareness, in line with some of the main aims of meditation practice.

6 Meditation also re-configures the relations between the DMN and other large-scale brain  
7 networks. For the control network, ample evidence suggests increased FC between the PCC  
8 and dorsolateral PFC both at rest and during meditation (64,72,73). Additionally, increased  
9 general FC between DMN and FPN during meditation (67) has been reported, but also  
10 increased DMN-FPN anti-correlations as a trait following meditation practice (67,74), which  
11 were associated with better performance in a sustained attention task (74). For the attention  
12 network, various studies found increased FC with the DMN following meditation practice (75–  
13 77), while another (78,79) reported stronger DAN-DMN anticorrelations in experienced  
14 mediators compared to controls during rest and sustained attention. These findings may  
15 suggest that meditation entails more directed resting-state mental processes, while reducing  
16 mind-wandering during attention-based tasks. Additionally, increased FC with the DMN during  
17 meditation and at rest has been reported both for the SAL (80,81) and somatosensory network  
18 (65,68,82). Taken together, these results suggest that as meditation practice proceeds, a  
19 reconfiguration of large-scale networks within the brain emerges, reflecting a greater capacity  
20 for sustained attention, meta-awareness and body-awareness, and a more fluid switching  
21 between internal and external attention, mediated by the SAL. We have highlighted the  
22 interactions among these networks across various attention and meditation states in **Figure 1**.

# 1 Trait Level Changes

2 We next highlight trait level changes for DMN connectivity across aging and various clinical  
 3 disorders. As individuals age, within-network DMN FC decreases but is accompanied by an  
 4 increase in across-network DMN-DAN FC, highlighting brain-wide network changes with aging  
 5 (83). Different developmental, neurodegenerative, and mood disorders impact the transmodal  
 6 networks centered on the DMN in varied ways. Here, we highlight the diverse literature across a  
 7 range of disorders to find common dysfunctional connectivity changes.

## 10 1. Neurodevelopmental Disorders:

### 11 a. Autism Spectrum Disorder (ASD)

12 ASD is associated with deficits in social communication and interaction as well as mentalizing to  
 13 infer mental states of others (82,83). Studies have demonstrated altered functional and  
 14 structural organization of the DMN in individuals with ASD, with atypical developmental  
 15 trajectories being prominent features (85). For example, most intrinsic FC studies in children  
 16 with ASD report increased within-network connectivity between core DMN nodes, while studies  
 17 in adolescents and adults report decreased connectivity, and studies in mixed age groups report  
 18 both increases and decreases (86–91). These inconsistencies likely reflect developmental  
 19 changes and heterogeneity in FC profiles across different nodes of the DMN. Nonetheless,  
 20 aberrancies in key nodes of the DMN and impairments in flexibly attending to socially relevant  
 21 stimuli have been generally observed, impacting social cognition and restricted and repetitive  
 22 behaviors in individuals with ASD. The strength of FC within DMN nodes has been associated  
 23 with autism spectrum traits, suggesting a link between DMN connectivity and the level of autism  
 24 symptomatology (85,92,93).

### 26 b. Attention Deficit Hyperactivity Disorder (ADHD)

The earliest study (94) to link DMN integrity to ADHD found that in a group of diagnosed adults, decreased anti-correlation was observed between the PCC node of the DMN and the ACC. Various fMRI studies have pointed to the enhancement of DMN activation in subjects with ADHD (95). Barring some diverse findings across task-based studies, the DMN is known to have an abnormal intrusion in the functioning of cognitive control network, leading to lapses in attention that are often observed in subjects with ADHD (96).

## 2. Neurological and Neurodegenerative Disorders:

### a. Alzheimer's disease (AD)

While the integrity of the DMN decreases with aging even in healthy subjects, FC of the DMN is found to be decreased in AD, especially in the PCC and precuneus (97). A systematic review showed that connectivity of DMN with SAL is increased in AD subjects (98) whereas the anticorrelation of DAN decreases with the DMN (99). Though some studies indicate FPN regions have increased FC with DMN (100), the findings in the literature are mixed (98).

### b. Parkinson's Disease (PD)

Studies of PD have found that DMN function and connectivity is altered, particularly in patients with cognitive impairments. FC within the DMN was found to be negatively correlated with cognitive composite z-scores from cognitive tests across multiple domains (memory, attention, language, executive and visuo-spatial abilities) in patients, suggesting that increase in within-DMN connectivity and failure to suppress it during executive tasks are associated with cognitive decline in PD (101–103). Another study focused on DMN dysfunction during an executive task and found PD patients exhibited less DMN deactivation during tasks that require externally-focused executive function (104).

### 3. Mood and Anxiety Disorders:

#### a. Major Depressive Disorder (MDD)

The DMN has long been known for its role in emotional activity and regulation of self-referential processes in MDD (105). The DMN shows increased intra-network as well as inter-network FC in MDD studies. More specifically, while the DMN is known to be anti-correlated with non-DMN networks like sensory and attentional networks, this anti-correlation is replaced by an abnormally positive correlation in MDD groups (106,107). Similar findings have been observed with respect to decreased anti-correlation of the DMN with the CEN/FPN in MDD groups (108). As well, MDD is also associated with a trait-level spatial expansion of the SAL that can infringe upon and shrink the DMN (109). Future work will be needed to understand how such alterations to spatial configuration impact DMN connectivity in MDD.

#### b. Schizophrenia (SZ)

SZ studies have found evidence of changes in FC between DMN and attention networks in the symptomatology of SZ groups (110–112). DMN connectivity in SZ is known to be increased between its midline hubs of mPFC and PCC (111,113,114), suggesting impaired self- and reality-monitoring. Moreover, increased FC between DMN and insula is well documented (115), and may relate to diminished internal/external switching in SZ, given the insula's role in mediation between the DMN and task-positive networks (4). In task paradigms, studies have found a stronger cue-induced DMN deactivation in a visuospatial attention task (116) and insufficient deactivation in a target detection task (117) in SZ groups. Impaired cognitive function in SZ is known to be linked with inefficient DMN suppression (118). Studies on audio-visual hallucinations further implicate the auditory cortices (i.e., superior and middle temporal gyri) based on their altered activation patterns during auditory processing (119) and altered FC with the DMN (120). It has been hypothesized that increased mPFC connectivity with the auditory

cortices in the superior temporal gyrus (STG) may contribute to source misattribution of self-generated auditory content as external stimuli (121).

The altered DMN FC observed across diverse clinical conditions mentioned here suggests a shared transdiagnostic dysregulation of the DMN. However, the pathophysiological mechanisms underlying these changes are likely distinct and warrant further investigation (122). This convergence raises the possibility of DMN FC as a broad biomarker of network dysfunction, although more work is needed to parse disorder-specific patterns. We have summarized the FC changes amongst the canonical networks for the various disorders in **Figure 2** and have covered additional disorders in **Supplemental Section 1**.

## fMRI Guidelines and Recommendations

Measuring between-network connectivity, such as DMN FC, crucially depends on the selection of network seeds and FC metrics—decision points that are hardly uniform across the field (123). Fortunately, methodological advances along with the hindsight of over two decades of fMRI research permit us to outline a set of recommended practices to guide future research.

### **Network estimation**

Network estimation is a data compression that transforms brain imaging data from the ‘microscale’ voxel to the ‘macroscale’ brain network. Many network estimation techniques are available today including dataset-derived parcellations (16,124), preset brain atlases (8,125–127), and functional mode atlases (128), each of which can be applied at the group- or subject-level. These different techniques have, unsurprisingly, a profound impact on FC results.

One of the greatest sources of variance introduced by network estimation is the obfuscation of meaningful subject-level signal structure that arises when averaging subject-level data in group-level analyses. Though group-level studies have long been the norm and provided many insights into the brain function and organization, there is growing recognition of the importance of subject-level precision fMRI given high structural and functional cross-subject spatial variance (1,10,129,130). Such cross-subject variance is highest in transmodal cortices including the DMN, FPN, and likely to a lesser extent, DAN (131,132). Strikingly, Bijsterbosch et al. (2018) found that up to 62% of the variance in simulated group-level FC network matrices is explained by cross-subject spatial variation in functional networks. This has profound implications for using group-level network FC as a biomarker.

Future research targeting DMN FC as biomarkers should face this cross-subject variance head-on by incorporating precision fMRI routines when possible. Through preserving meaningful signal structure and allowing for more accurate modeling of variance (130), subject-level approaches would improve effect detection and interpretative clarity. Such an approach would be paramount, for instance, to understanding the differential contribution of DMN and FPN subsystems for FC biomarkers (see, for instance:(133)).

Barriers to precision fMRI include collecting enough data per participant (13,130,134) and the difficulty of group-level inference. Fortunately, advancements in fMRI acquisition with multi-echo sequences (133,135) and individualized areal estimation (16,136) are lowering the data collection barrier, while individualized-but-generalized parcellations (137) and forthcoming functional alignment procedures (137) may help allow broader inference. When precision approaches are not feasible, we recommend researchers compare results across network maps, such as with atlas-comparison tools (126,138).

## Functional Connectivity estimation

FC is perhaps the most popular metric used to compute inter-regional communication in fMRI. It is also, somewhat circularly, inseparable from most network estimation methods (129,139).

While a detailed review of FC estimation methods is beyond the scope of this review (see (123)), certain practices have immediate relevance to measuring between-network FC.

One such practice is global signal regression (GSR), where the ‘global’ average time series is removed through linear regression. GSR has long been controversial in fMRI research due to its uncertain nature (140), non-uniformity (141), and profound effect on FC estimates including anticorrelations (142). An emerging understanding of the global signal is that it represents a composite of both neural and non-neural sources (140,143), where removal of the global signal increases anticorrelations between brain systems by virtue of mean centering (142). In some cases, omitting GSR, particularly for subjects with large variation in this signal due to irregular respiration, can dramatically alter FC effects. For example, a recent meta-analysis of TMS targeting outcomes reported that associations between treatment response and subgenual ACC FC are driven by patients with strong global signal fluctuations (144). This suggests that the use of GSR can influence our ability to detect clinically relevant DMN FC patterns. Effects that are not reproducible without GSR may not be analysis artifacts in as much as they reflect neural features whose detection benefits from removal of shared trends (142). In general, and especially for research focused on brain system anticorrelations, we recommend reporting results both with and without GSR. Recent noise cleaning methods improve noise based correlations.

Beyond GSR, other preprocessing techniques—such as motion correction, band-pass filtering, and spatial smoothing also play critical roles in shaping FC results. Standardizing these

1 methods across studies could improve the reliability and reproducibility of DMN connectivity  
2 findings like recent studies have attempted (145), enhancing its potential as a biomarker for  
3 disease and treatment outcomes. We recommend the use of standardized preprocessing  
4 pipelines tools like fMRIPrep (146).

5  
6 Alternative metrics for inter-regional communication should also be explored. For instance, the  
7 bivariate correlation underlying most FC approaches measures only a subset of the information  
8 architecture of the brain and even BOLD signal (139,147), with particular relevance for  
9 transmodal networks (148). While FC reflects redundant information equally carried by multiple  
10 sources, partial entropy-related measures can compute synergistic information that is  
11 necessarily supported by multiple sources (139,147).

## 12 Conclusions and Future Directions

13 In the current review, we summarized studies of FC between DMN and other brain regions  
14 during various states including attentional processes, mind wandering and meditation and  
15 highlighted how traits change with aging and various clinical conditions. We concluded with  
16 general guidelines on how to robustly measure this relationship. Overall, the DMN allows for  
17 elaborate scene construction, episodic projection, internal mentation and thinking about others  
18 and also involved in episodic memory retrieval, semantic processing, goal directed cognition.  
19 Disorders like depression, schizophrenia, and ADHD feature marked alterations to DMN activity  
20 and connectivity with control and attention networks. Such changes appear linked to reduced  
21 control over intrinsic activity as well as reduced ability to switch between extrinsic and intrinsic  
22 modes of processing. The ease of collecting FC data gives us a potential utility of DMN  
23 connectivity as a biomarker of mental health and disorders and how therapeutic interventions  
24 including meditation paradigms allow the maintenance of DMN health. More work is needed on  
25 making the DMN connectivity measure robust across individual healthy subjects and increasing



its clinical viability, which would require a clear definition of networks and where precision imaging approaches will help. The development of standardized collection and preprocessing frameworks will also be required.

As we move forward, several avenues warrant further exploration to deepen our understanding of the DMN and its implications as a biomarker for cognitive health: a) Precision fMRI Methods: Embracing subject-level precision fMRI approaches will enable researchers to better capture individual variability in DMN connectivity, enhancing the reliability and interpretability of findings. b) Advanced Connectivity Metrics: Investigating alternative metrics beyond traditional BOLD FC, such as partial entropy-related measures, may unveil novel insights into the complex dynamics of DMN interactions. c) Analytical approach for examining current findings: a meta-analysis of discrepancies in literature, focusing on variations in processing techniques, network estimation, FC methods etc. d) Integration of Multimodal Imaging: Combining fMRI with other modalities like EEG-fMRI or MEG can provide a more comprehensive understanding of the neural bases underlying DMN connectivity, offering valuable insights into brain-behavior relationships (149). e) Clinical Translation and Intervention: Translating research findings into clinical practice, including the development of targeted interventions leveraging DMN modulation, holds promise for improving treatment outcomes in psychiatric disorders. f) Predictive modeling techniques like CPM (40) and multivariate approaches (150) would be useful in establishing the reliability of DMN FC as a biomarker across states, traits and disorders. g) Longitudinal Studies and Biomarker Development: Longitudinal studies tracking DMN connectivity changes over time, coupled with biomarker development efforts would be crucial for elucidating disease trajectories and identifying early markers of cognitive decline or psychiatric vulnerability h) Mega-Analysis studies like the ENIGMA consortiums to aid in large-scale development and validation of DMN biomarkers.

1 By embracing these future directions, we could unlock new frontiers in DMN research,  
2 advancing our knowledge of brain function and facilitating the development of innovative  
3 strategies for mental health diagnosis and intervention.  
4  
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## Figure Captions

**Figure 1:** a) Top panel highlights the nodes of the DMN and other large-scale brain networks: DAN, FPN and SAL as defined using the 7-Network Yeo 2011 atlas. Bottom panel depicts the time series of the four networks in a resting-state condition for an example subject. The connectivity dynamically changes across the network's time. b) A depiction of the resting state functional connectivity matrix averaged across subjects for the four networks using the Schaefer 100 parcellation scheme. We see strong within-network connectivity but also within and across network heterogeneity highlighting the need of a common taxonomy and procedures for the estimation of these networks. Functional connectivity summary plots for the four networks for c) fixation based resting state (as computed on healthy subjects from the Human Connectome Project dataset as depicted in panel b) and for attention and meditation states (as observed from literature, see supplementary table 1 for reference to specific edges). d) Summary plots for goal directed cognition, idea generation and evaluation cognitive states which highlights DMN-FPN working together. Red line represents positive correlation and the blue line represents negative correlation (anti-correlation) within/between the networks. Dashed line highlights mixed results in the literature regarding the strength of the connectivity but more biased towards the highlighted color. Self loops indicate within network connectivity.

**Figure 2:** Highlighting the changes in the connectivity within and across the four networks: DMN, DAN, FPN and SAL across mental disorders with most consistent results as mentioned in the literature (see supplementary table 2 for reference to specific edges) a) Neurodegenerative disorders: Parkinson's Disease (PD), Alzheimer's Disease (AD), Multiple Sclerosis (MS); b) Mood and anxiety disorders: Mild Depressive Disorder (MDD), Post Traumatic Stress Disorder (PTSD), Obsessive Control Disorder (OCD); c) Developmental disorders: Attention Deficit Hyperactivity Disorder (ADHD), Bipolar Disorder (BD) and Schizophrenia (SZ). Green line/loop

represents an increase in the amplitude of connectivity whereas orange line/loop represents decreased connectivity and the dashed line indicates mixed results in the literature.

## Disclosures

AZ is supported by PJT-168974 from CIHR and RGPIN-2018-04293 from NSERC. LQU is supported by R21HD111805 from NICHD and U01DA050987 from NIDA. RSP is supported by grants R01AG054427 and R61AG081982 from NIA of the NIH.

The authors report no biomedical financial interests or potential conflicts of interest.

## Supplement Description:

Supplement Text and Tables S1-S2

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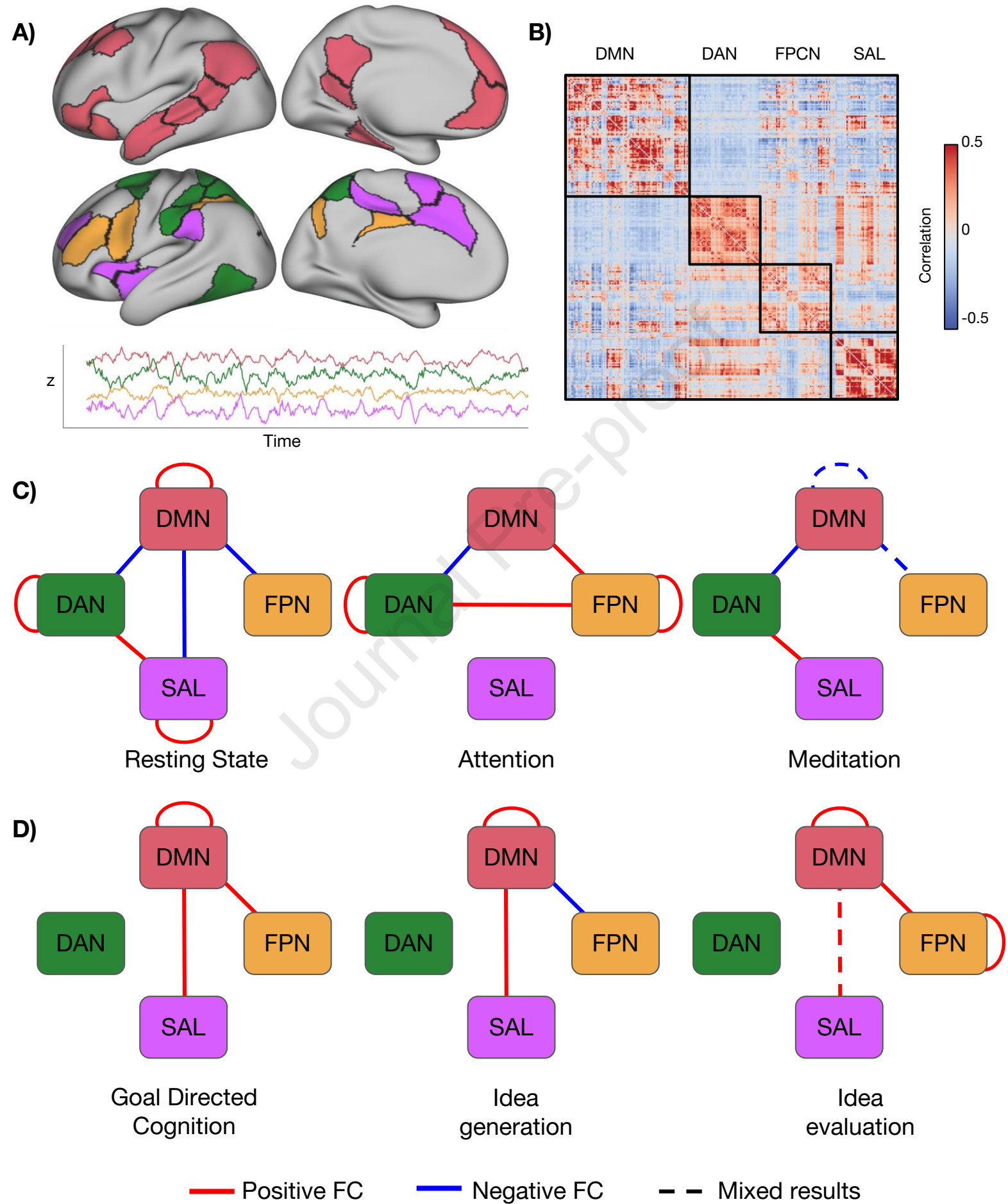


Figure 1

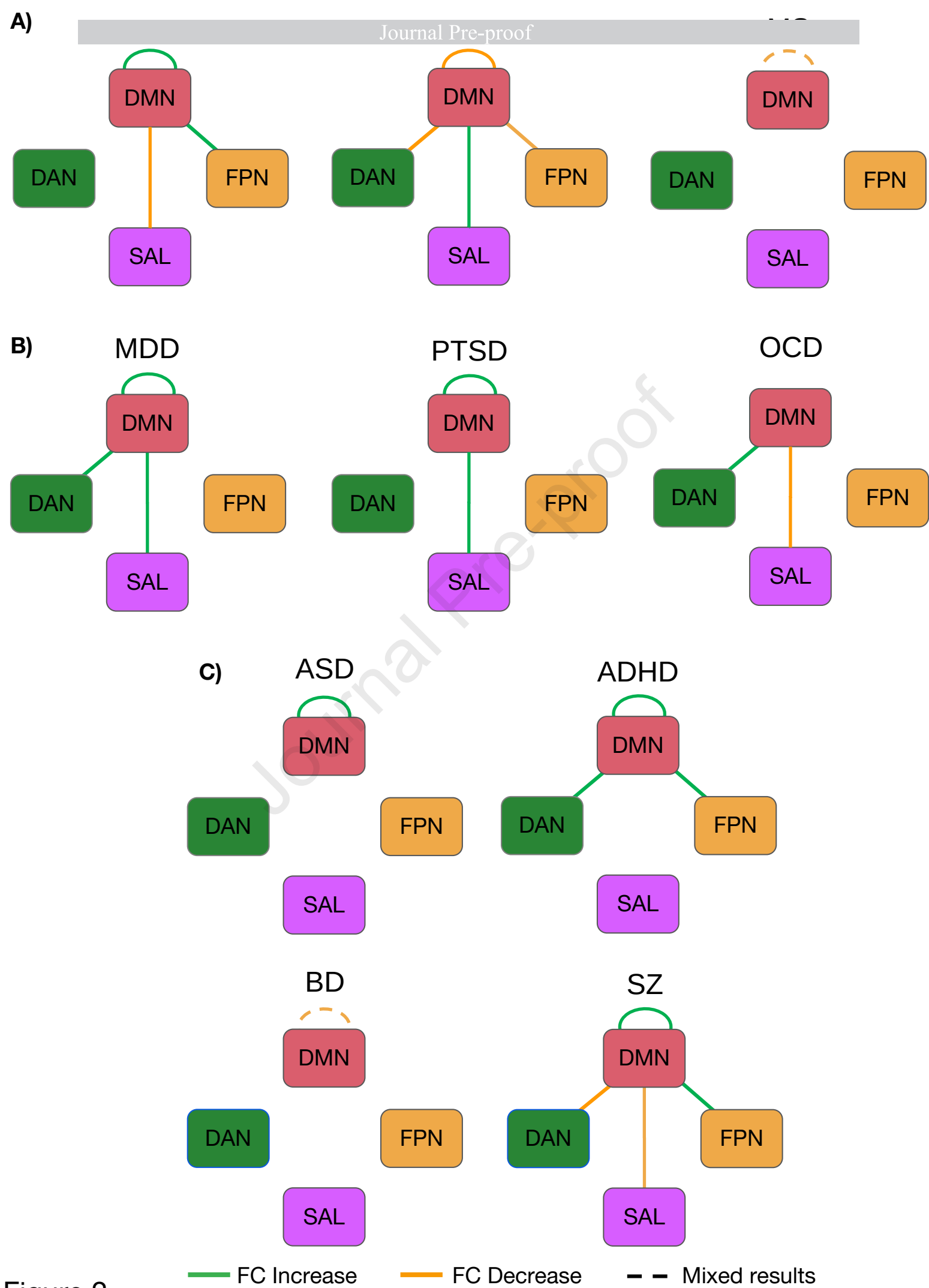


Figure 2